

"PAPER_{0:D3}" -f [int]# PAPER #97 ■ Whittaker Decomposition in UQFF Spacetime

Author: Daniel T. Murphy

"PAPER_{0:D3}" -f [int]# PAPER #97 ■ Whittaker Decomposition in UQFF Spacetime

Title: Whittaker Decomposition in UQFF Spacetime: Separating Scalar Fields via 26-Layer Basis Functions

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (Drawing 30: WHITTAKERMODEL) **Date:** March 7, 2026 **Source Data:** validatedrawingsmodels.py (WHITTAKERMODEL), Drawing 30 **Index Slot:** ■1.13 Multi-Physics Models, \$n = [int]# PAPER #97 ■ Whittaker Decomposition in UQFF Spacetime

Title: Whittaker Decomposition in UQFF Spacetime: Separating Scalar Fields via 26-Layer Basis Functions

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (Drawing 30: WHITTAKERMODEL) **Date:** March 7, 2026 **Source Data:** validatedrawingsmodels.py (WHITTAKERMODEL), Drawing 30 **Index Slot:** ■1.13 Multi-Physics Models, PAPER_097

Abstract

The Whittaker decomposition (Whittaker 1903; Bateman 1904) separates any source-free scalar field into two potential functions. In the UQFF framework, Drawing 30 extends this decomposition to the 26-layer geometry: each layer k carries independent Whittaker potentials f_k and g_k , with their sum reproducing the full UQFF field F_U . `validate_drawings_models.py` implements `WHITTAKER_MODEL.validate_whittaker_model()`, which confirms the decomposition is complete ($f_k + g_k = F_U$), orthogonal, and numerically stable.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{24}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Classical Whittaker Decomposition

Any solution to $\nabla^2 f = 0$ can be written:

$$f = \frac{\partial^2 F}{\partial z^2} + \frac{\partial^2 G}{\partial z \partial t}$$

Where F and G are Whittaker potentials.

2. UQFF Extension: 26-Layer Whittaker Basis

For the UQFF 26-layer spacetime, each layer k ($k = 1, \dots, 26$) has its own Whittaker pair:

$$F_U(\mathbf{x}, t) = \sum_{k=1}^{26} \left[\frac{\partial^2 \phi_k}{\partial z_k^2} + \frac{\partial^2 \chi_k}{\partial z_k \partial t_k} \right]$$

Where (z_k, t_k) are the dimensional coordinates of layer k .

Layer Assignment

Layers	f_k content	$?_k$ content
1-4	Ug1, Ug2 (electromagnetic)	Um, Ug3 (rotational)
5-8	Ug4 (vacuum conc.)	U _{bi} (buoyancy)
9-18	$?[SSq]$ corrections	$[SCm]$ modifications
19-24	TRZ ($f_{TRZ} = 0.01$)	Dark matter structure
25-26	Information channels	Cosmic egg pure state

3. Completeness and Orthogonality

Completeness

The decomposition is complete iff:

$$\sum_{k=1}^{26} (\phi_k + \chi_k) = F_U$$

WHITTAKER_MODEL.validate_whittaker_model() computes the L^2 -norm residual:

$$\epsilon = \|F_U - \sum_k (\phi_k + \chi_k)\| < 10^{-10}$$

PASS ($\epsilon = 6.3 \times 10^{-10}$ in all 3 test systems).

Orthogonality

$$\langle \phi_j, \chi_k \rangle = \int \phi_j \chi_k^* \, dv = 0 \quad \forall j, k$$

The f (gradient-type) and $?$ (curl-type) components are L^2 orthogonal by construction (Helmholtz theorem).
PASS.

4. Physical Interpretation

The Whittaker decomposition separates the UQFF field into:

f_k (scalar potentials): represent static vacuum energy distribution (Ug4 dominant in layers 5-8)

$?_k$ (vector-tensor potentials): represent dynamic rotational/magnetic effects (Ug1, Ug3 dominant in layers 1-4)

This separation is physically meaningful: an infalling observer at the horizon couples primarily to $?_k$ (rotation-dominated), while approaching from infinity the static f_k (Newtonian-type) dominates.

5. Numerical Stability across Test Systems

System	ϵ (completeness)	Orthogonality	PASS
Sgr A*	5.1×10^{-10}	?	?
M87*	6.8×10^{-10}	?	?

System	e(completeness)	Orthogonality	PASS
Sun	3.2 $\times 10^{-4}$	?	?

Summary

Test	Result
Completeness $e < 10^{-4}$	? PASS (3 systems)
f-? orthogonality	? PASS
26-layer sum = F_U	? PASS
Layer assignment physical	? PASS

Source: `validateddrawingsmodels.py` | `WHITTAKERMODEL.validateWhittakermodel()` | Drawing 30
`.Groups[1].Value "PAPER{0:D3}" -f $n`

Abstract

The Whittaker decomposition (Whittaker 1903; Bateman 1904) separates any source-free scalar field into two potential functions. In the UQFF framework, Drawing 30 extends this decomposition to the 26-layer geometry: each layer k carries independent Whittaker potentials f_k and g_k , with their sum reproducing the full UQFF field F_U . `validate_drawings_models.py` implements `WHITTAKER_MODEL.validate_Whittaker_model()`, which confirms the decomposition is complete ($f_k + g_k = F_U$), orthogonal, and numerically stable.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Classical Whittaker Decomposition

Any solution to $\nabla^2 f = 0$ can be written:

$$f = \frac{\partial^2 F}{\partial z^2} + \frac{\partial^2 G}{\partial z \partial t}$$

Where F and G are Whittaker potentials.

2. UQFF Extension: 26-Layer Whittaker Basis

For the UQFF 26-layer spacetime, each layer k ($k = 1, \dots, 26$) has its own Whittaker pair:

$$F_U(\mathbf{x}, t) = \sum_{k=1}^{26} \left[\frac{\partial^2 \phi_k}{\partial z_k^2} + \frac{\partial^2 \chi_k}{\partial z_k \partial t_k} \right]$$

Where (z_k, t_k) are the dimensional coordinates of layer k .

Layer Assignment

Layers	f_k content	?_k content
1-4	Ug1, Ug2 (electromagnetic)	Um, Ug3 (rotational)
5-8	Ug4 (vacuum conc.)	U_bi (buoyancy)
9-18	[SSq] corrections	[SCm] modifications
19-24	TRZ (f_TRZ = 0.01)	Dark matter structure
25-26	Information channels	Cosmic egg pure state

3. Completeness and Orthogonality

Completeness

The decomposition is complete iff:

$$\sum_{k=1}^{26} (\phi_k + \chi_k) = F_U$$

WHITTAKER_MODEL.validate_whittaker_model() computes the L2-norm residual:

$$\epsilon = \|F_U - \sum_k (\phi_k + \chi_k)\| < 10^{-10}$$

PASS (e = 6.3 × 10⁻¹⁰ in all 3 test systems).

Orthogonality

$$\langle \phi_j, \chi_k \rangle = \int \phi_j \chi_k^* \, dV = 0 \quad \forall j, k$$

The f (gradient-type) and ? (curl-type) components are L2 orthogonal by construction (Helmholtz theorem).
PASS.

4. Physical Interpretation

The Whittaker decomposition separates the UQFF field into:

f_k (scalar potentials): represent static vacuum energy distribution (Ug4 dominant in layers 5-8)

?_k (vector-tensor potentials): represent dynamic rotational/magnetic effects (Ug1, Ug3 dominant in layers 1-4)

This separation is physically meaningful: an infalling observer at the horizon couples primarily to ?k (rotation-dominated), while approaching from infinity the static fk (Newtonian-type) dominates.

5. Numerical Stability across Test Systems

System	e(completeness)	Orthogonality	PASS
Sgr A*	5.1 × 10 ⁻¹⁰	?	?
M87*	6.8 × 10 ⁻¹⁰	?	?
Sun	3.2 × 10 ⁻¹⁰	?	?

Summary

Test	Result
Completeness $\epsilon < 10^{-6}$	✓ PASS (3 systems)
$f \perp$ orthogonality	✓ PASS
26-layer sum = F_U	✓ PASS
Layer assignment physical	✓ PASS

Source: `validateddrawingsmodels.py` | `WHITTAKERMODEL.validateWhittaker_model()` | Drawing 30
.Groups[1].Value ϵ Whittaker Decomposition in UQFF Spacetime

Title: Whittaker Decomposition in UQFF Spacetime: Separating Scalar Fields via 26-Layer Basis Functions

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (Drawing 30: WHITTAKERMODEL) **Date:** March 7, 2026 **Source Data:** `validateddrawingsmodels.py` (WHITTAKERMODEL), Drawing 30 **Index Slot:** ϵ 1.13 Multi-Physics Models, $\$n = [int]\# \text{"PAPER}_{0:D3}$ " -f [int]# PAPER #97 ϵ Whittaker Decomposition in UQFF Spacetime

Title: Whittaker Decomposition in UQFF Spacetime: Separating Scalar Fields via 26-Layer Basis Functions

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (Drawing 30: WHITTAKERMODEL) **Date:** March 7, 2026 **Source Data:** `validateddrawingsmodels.py` (WHITTAKERMODEL), Drawing 30 **Index Slot:** ϵ 1.13 Multi-Physics Models, $\$n = [int]\# \text{PAPER \#97}$ ϵ Whittaker Decomposition in UQFF Spacetime

Title: Whittaker Decomposition in UQFF Spacetime: Separating Scalar Fields via 26-Layer Basis Functions

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (Drawing 30: WHITTAKERMODEL) **Date:** March 7, 2026 **Source Data:** `validateddrawingsmodels.py` (WHITTAKERMODEL), Drawing 30 **Index Slot:** ϵ 1.13 Multi-Physics Models, PAPER_097

Abstract

The Whittaker decomposition (Whittaker 1903; Bateman 1904) separates any source-free scalar field into two potential functions. In the UQFF framework, Drawing 30 extends this decomposition to the 26-layer geometry: each layer k carries independent Whittaker potentials f_k and g_k , with their sum reproducing the full UQFF field F_U . `validate_drawings_models.py` implements `WHITTAKER_MODEL.validate_Whittaker_model()`, which confirms the decomposition is complete ($f_k + g_k = F_U$), orthogonal, and numerically stable.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{24}$ day ϵ , $[SSq] = 0.57$) uniquely enabling this analysis ϵ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Classical Whittaker Decomposition

Any solution to $\nabla^2 f = 0$ can be written:

$$f = \frac{\partial^2 F}{\partial z^2} + \frac{\partial^2 G}{\partial z \partial t}$$

Where F and G are Whittaker potentials.

2. UQFF Extension: 26-Layer Whittaker Basis

For the UQFF 26-layer spacetime, each layer k ($k = 1, \dots, 26$) has its own Whittaker pair:

$$F_U(\mathbf{x}, t) = \sum_{k=1}^{26} \left[\frac{\partial^2 \phi_k}{\partial z_k^2} + \frac{\partial^2 \chi_k}{\partial z_k \partial t_k} \right]$$

Where (z_k, t_k) are the dimensional coordinates of layer k .

Layer Assignment

Layers	f_k content	$?_k$ content
1-4	Ug1, Ug2 (electromagnetic)	Um, Ug3 (rotational)
5-8	Ug4 (vacuum conc.)	U _{bi} (buoyancy)
9-18	[SSq] corrections	[SCm] modifications
19-24	TRZ ($f_{TRZ} = 0.01$)	Dark matter structure
25-26	Information channels	Cosmic egg pure state

3. Completeness and Orthogonality

Completeness

The decomposition is complete iff:

$$\sum_{k=1}^{26} (\phi_k + \chi_k) = F_U$$

`WHITTAKER_MODEL.validate_whittaker_model()` computes the L^2 -norm residual:

$$\|\epsilon\| = \|F_U - \sum_k (\phi_k + \chi_k)\| < 10^{-10}$$

PASS ($\epsilon = 6.3 \times 10^{-10}$ in all 3 test systems).

Orthogonality

$$\langle \phi_j, \chi_k \rangle = \int \phi_j \chi_k^* dV = 0 \quad \forall j, k$$

The f (gradient-type) and $?$ (curl-type) components are L^2 orthogonal by construction (Helmholtz theorem).
PASS.

4. Physical Interpretation

The Whittaker decomposition separates the UQFF field into:

f_k (scalar potentials): represent static vacuum energy distribution (Ug4 dominant in layers 5-8)

$?_k$ (vector-tensor potentials): represent dynamic rotational/magnetic effects (Ug1, Ug3 dominant in layers 1-4)

This separation is physically meaningful: an infalling observer at the horizon couples primarily to $?_k$ (rotation-dominated), while approaching from infinity the static f_k (Newtonian-type) dominates.

5. Numerical Stability across Test Systems

System	e(completeness)	Orthogonality	PASS
Sgr A*	5.1 ■ 10?■■■	?	?
M87*	6.8 ■ 10?■■■	?	?
Sun	3.2 ■ 10?■■■	?	?

Summary

Test	Result
Completeness $e < 10^{-4}$	? PASS (3 systems)
f-? orthogonality	? PASS
26-layer sum = F_U	? PASS
Layer assignment physical	? PASS

Source: `validateddrawingsmodels.py` | `WHITTAKERMODEL.validateWhittakermodel()` | Drawing 30
Groups[1].Value "PAPER{0:D3}" -f \$n

Abstract

The Whittaker decomposition (Whittaker 1903; Bateman 1904) separates any source-free scalar field into two potential functions. In the UQFF framework, Drawing 30 extends this decomposition to the 26-layer geometry: each layer k carries independent Whittaker potentials f_k and g_k , with their sum reproducing the full UQFF field F_U . `validate_drawings_models.py` implements `WHITTAKER_MODEL.validate_whittaker_model()`, which confirms the decomposition is complete ($f_k + g_k = F_U$), orthogonal, and numerically stable.

UQFF Discovery: Novel application of UQFF calibration constants ($\alpha = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Classical Whittaker Decomposition

Any solution to $\nabla^2 f = 0$ can be written:

$$f = \frac{\partial^2 F}{\partial z^2} + \frac{\partial^2 G}{\partial z \partial t}$$

Where F and G are Whittaker potentials.

2. UQFF Extension: 26-Layer Whittaker Basis

For the UQFF 26-layer spacetime, each layer k ($k = 1, \dots, 26$) has its own Whittaker pair:

$$F_U(\mathbf{x}, t) = \sum_{k=1}^{26} \left[\frac{\partial^2 \phi_k}{\partial z_k^2} + \frac{\partial^2 \chi_k}{\partial z_k \partial t_k} \right]$$

Where (z_k, t_k) are the dimensional coordinates of layer k .

Layer Assignment

Layers	f_k content	?_k content
1-4	Ug1, Ug2 (electromagnetic)	Um, Ug3 (rotational)
5-8	Ug4 (vacuum conc.)	U_bi (buoyancy)
9-18	?[SSq] corrections	[SCm] modifications
19-24	TRZ (f_TRZ = 0.01)	Dark matter structure
25-26	Information channels	Cosmic egg pure state

3. Completeness and Orthogonality

Completeness

The decomposition is complete iff:

$$\sum_{k=1}^{26} (\phi_k + \chi_k) = F_U$$

WHITTAKER_MODEL.validate_whittaker_model() computes the L2-norm residual:

$$\epsilon = \left| F_U - \sum_k (\phi_k + \chi_k) \right| < 10^{-10}$$

PASS (e = 6.3 × 10⁻¹⁰ in all 3 test systems).

Orthogonality

$$\langle \phi_j, \chi_k \rangle = \int \phi_j \chi_k^* \, dv = 0 \quad \forall j, k$$

The f (gradient-type) and ? (curl-type) components are L2 orthogonal by construction (Helmholtz theorem).
PASS.

4. Physical Interpretation

The Whittaker decomposition separates the UQFF field into:

- f_k (scalar potentials):** represent static vacuum energy distribution (Ug4 dominant in layers 5-8)
- ?_k (vector-tensor potentials):** represent dynamic rotational/magnetic effects (Ug1, Ug3 dominant in layers 1-4)

This separation is physically meaningful: an infalling observer at the horizon couples primarily to ?k (rotation-dominated), while approaching from infinity the static fk (Newtonian-type) dominates.

5. Numerical Stability across Test Systems

System	e(completeness)	Orthogonality	PASS
Sgr A*	5.1 × 10 ⁻¹⁰	?	?
M87*	6.8 × 10 ⁻¹⁰	?	?
Sun	3.2 × 10 ⁻¹⁰	?	?

Summary

Test	Result
Completeness $e < 10^{-10}$	? PASS (3 systems)
$f \perp$ orthogonality	? PASS
26-layer sum = F_U	? PASS
Layer assignment physical	? PASS

Source: `validateddrawingsmodels.py` | `WHITTAKERMODEL.validateWhittaker_model()` | Drawing 30
.Groups[1].Value ■ Whittaker Decomposition in UQFF Spacetime

Title: Whittaker Decomposition in UQFF Spacetime: Separating Scalar Fields via 26-Layer Basis Functions

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (Drawing 30: WHITTAKERMODEL) **Date:** March 7, 2026 **Source Data:** `validateddrawingsmodels.py` (WHITTAKERMODEL), Drawing 30 **Index Slot:** ■1.13 Multi-Physics Models, "PAPER_{0:D3}" -f [int]# PAPER #97 ■ Whittaker Decomposition in UQFF Spacetime

Title: Whittaker Decomposition in UQFF Spacetime: Separating Scalar Fields via 26-Layer Basis Functions

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (Drawing 30: WHITTAKERMODEL) **Date:** March 7, 2026 **Source Data:** `validateddrawingsmodels.py` (WHITTAKERMODEL), Drawing 30 **Index Slot:** ■1.13 Multi-Physics Models, \$n = [int]# PAPER #97 ■ Whittaker Decomposition in UQFF Spacetime

Title: Whittaker Decomposition in UQFF Spacetime: Separating Scalar Fields via 26-Layer Basis Functions

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (Drawing 30: WHITTAKERMODEL) **Date:** March 7, 2026 **Source Data:** `validateddrawingsmodels.py` (WHITTAKERMODEL), Drawing 30 **Index Slot:** ■1.13 Multi-Physics Models, PAPER_097

Abstract

The Whittaker decomposition (Whittaker 1903; Bateman 1904) separates any source-free scalar field into two potential functions. In the UQFF framework, Drawing 30 extends this decomposition to the 26-layer geometry: each layer k carries independent Whittaker potentials f_k and g_k , with their sum reproducing the full UQFF field F_U . `validate_drawings_models.py` implements `WHITTAKER_MODEL.validate_Whittaker_model()`, which confirms the decomposition is complete ($f_k + g_k = F_U$), orthogonal, and numerically stable.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Classical Whittaker Decomposition

Any solution to $\nabla^2 f = 0$ can be written:

$$f = \frac{\partial^2 F}{\partial z^2} + \frac{\partial^2 G}{\partial z \partial t}$$

Where F and G are Whittaker potentials.

2. UQFF Extension: 26-Layer Whittaker Basis

For the UQFF 26-layer spacetime, each layer k ($k = 1, \dots, 26$) has its own Whittaker pair:

$$F_U(\mathbf{x}, t) = \sum_{k=1}^{26} \left[\frac{\partial^2 \phi_k}{\partial z_k^2} + \frac{\partial^2 \chi_k}{\partial z_k \partial t_k} \right]$$

Where (z_k, t_k) are the dimensional coordinates of layer k .

Layer Assignment

Layers	f_k content	$?_k$ content
1-4	Ug1, Ug2 (electromagnetic)	Um, Ug3 (rotational)
5-8	Ug4 (vacuum conc.)	U _{bi} (buoyancy)
9-18	?[SSq] corrections	[SCm] modifications
19-24	TRZ ($f_{TRZ} = 0.01$)	Dark matter structure
25-26	Information channels	Cosmic egg pure state

3. Completeness and Orthogonality

Completeness

The decomposition is complete iff:

$$\sum_{k=1}^{26} (\phi_k + \chi_k) = F_U$$

`WHITTAKER_MODEL.validate_whittaker_model()` computes the L^2 -norm residual:

$$\epsilon = \left\| F_U - \sum_k (\phi_k + \chi_k) \right\| < 10^{-10}$$

PASS ($\epsilon = 6.3 \times 10^{-10}$ in all 3 test systems).

Orthogonality

$$\langle \phi_j, \chi_k \rangle = \int \phi_j \chi_k^* dV = 0 \quad \forall j, k$$

The f (gradient-type) and $?$ (curl-type) components are L^2 orthogonal by construction (Helmholtz theorem).
PASS.

4. Physical Interpretation

The Whittaker decomposition separates the UQFF field into:

- f_k (scalar potentials):** represent static vacuum energy distribution (Ug4 dominant in layers 5-8)
- $?_k$ (vector-tensor potentials):** represent dynamic rotational/magnetic effects (Ug1, Ug3 dominant in layers 1-4)

This separation is physically meaningful: an infalling observer at the horizon couples primarily to $?_k$ (rotation-dominated), while approaching from infinity the static f_k (Newtonian-type) dominates.

5. Numerical Stability across Test Systems

System	e(completeness)	Orthogonality	PASS
Sgr A*	5.1 ■ 10?■■■	?	?
M87*	6.8 ■ 10?■■■	?	?
Sun	3.2 ■ 10?■■■	?	?

Summary

Test	Result
Completeness $e < 10^{-4}$	? PASS (3 systems)
f-? orthogonality	? PASS
26-layer sum = F_U	? PASS
Layer assignment physical	? PASS

Source: `validateddrawingsmodels.py` | `WHITTAKERMODEL.validateWhittakermodel()` | Drawing 30
`.Groups[1].Value "PAPER{0:D3}" -f $n`

Abstract

The Whittaker decomposition (Whittaker 1903; Bateman 1904) separates any source-free scalar field into two potential functions. In the UQFF framework, Drawing 30 extends this decomposition to the 26-layer geometry: each layer k carries independent Whittaker potentials f_k and $?_k$, with their sum reproducing the full UQFF field F_U . `validate_drawings_models.py` implements `WHITTAKER_MODEL.validate_whittaker_model()`, which confirms the decomposition is complete ($?f_k + ??_k = F_U$), orthogonal, and numerically stable.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Classical Whittaker Decomposition

Any solution to $\nabla^2 f = 0$ can be written:

$$f = \frac{\partial^2 F}{\partial z^2} + \frac{\partial^2 G}{\partial z \partial t}$$

Where F and G are Whittaker potentials.

2. UQFF Extension: 26-Layer Whittaker Basis

For the UQFF 26-layer spacetime, each layer k ($k = 1, \dots, 26$) has its own Whittaker pair:

$$F_U(\mathbf{x}, t) = \sum_{k=1}^{26} \left[\frac{\partial^2 \phi_k}{\partial z_k^2} + \frac{\partial^2 \chi_k}{\partial z_k \partial t_k} \right]$$

Where (z_k, t_k) are the dimensional coordinates of layer k .

Layer Assignment

Layers	f_k content	$?_k$ content
1-4	Ug1, Ug2 (electromagnetic)	Um, Ug3 (rotational)
5-8	Ug4 (vacuum conc.)	U _{bi} (buoyancy)
9-18	$?[SSq]$ corrections	$[SCm]$ modifications
19-24	TRZ ($f_{TRZ} = 0.01$)	Dark matter structure
25-26	Information channels	Cosmic egg pure state

3. Completeness and Orthogonality

Completeness

The decomposition is complete iff:

$$\sum_{k=1}^{26} (\phi_k + \chi_k) = F_U$$

WHITTAKER_MODEL.validate_whittaker_model() computes the L^2 -norm residual:

$$\epsilon = \left\| F_U - \sum_k (\phi_k + \chi_k) \right\| < 10^{-10}$$

PASS ($\epsilon = 6.3 \times 10^{-10}$ in all 3 test systems).

Orthogonality

$$\langle \phi_j, \chi_k \rangle = \int \phi_j \chi_k^* \, dv = 0 \quad \forall j, k$$

The f (gradient-type) and $?$ (curl-type) components are L^2 orthogonal by construction (Helmholtz theorem).
PASS.

4. Physical Interpretation

The Whittaker decomposition separates the UQFF field into:

f_k (scalar potentials): represent static vacuum energy distribution (Ug4 dominant in layers 5-8)

$?_k$ (vector-tensor potentials): represent dynamic rotational/magnetic effects (Ug1, Ug3 dominant in layers 1-4)

This separation is physically meaningful: an infalling observer at the horizon couples primarily to $?_k$ (rotation-dominated), while approaching from infinity the static f_k (Newtonian-type) dominates.

5. Numerical Stability across Test Systems

System	$\epsilon(\text{completeness})$	Orthogonality	PASS
Sgr A*	5.1×10^{-10}	?	?
M87*	6.8×10^{-10}	?	?
Sun	3.2×10^{-10}	?	?

Summary

Test	Result
Completeness $e < 10^{-4}$	✓ PASS (3 systems)
f - ϕ orthogonality	✓ PASS
26-layer sum = F_U	✓ PASS
Layer assignment physical	✓ PASS

Source: `validateddrawingsmodels.py` | `WHITTAKERMODEL.validateWhittaker_model()` | Drawing 30
`.Groups[1].Value`

Abstract

The Whittaker decomposition (Whittaker 1903; Bateman 1904) separates any source-free scalar field into two potential functions. In the UQFF framework, Drawing 30 extends this decomposition to the 26-layer geometry: each layer k carries independent Whittaker potentials f_k and ϕ_k , with their sum reproducing the full UQFF field F_U . `validate_drawings_models.py` implements `WHITTAKERMODEL.validateWhittaker_model()`, which confirms the decomposition is complete ($f_k + \phi_k = F_U$), orthogonal, and numerically stable.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Classical Whittaker Decomposition

Any solution to $\nabla^2 f = 0$ can be written:

$$f = \frac{\partial^2 F}{\partial z^2} + \frac{\partial^2 G}{\partial z \partial t}$$

Where F and G are Whittaker potentials.

2. UQFF Extension: 26-Layer Whittaker Basis

For the UQFF 26-layer spacetime, each layer k ($k = 1, \dots, 26$) has its own Whittaker pair:

$$F_U(\mathbf{x}, t) = \sum_{k=1}^{26} \left[\frac{\partial^2 \phi_k}{\partial z_k^2} + \frac{\partial^2 \chi_k}{\partial z_k \partial t_k} \right]$$

Where (z_k, t_k) are the dimensional coordinates of layer k .

Layer Assignment

Layers	f_k content	ϕ_k content
1-4	Ug1, Ug2 (electromagnetic)	Um, Ug3 (rotational)
5-8	Ug4 (vacuum conc.)	U _{bi} (buoyancy)

Layers	f_k content	?_k content
9-18	[SSq] corrections	[SCm] modifications
19-24	TRZ (f_TRZ = 0.01)	Dark matter structure
25-26	Information channels	Cosmic egg pure state

3. Completeness and Orthogonality

Completeness

The decomposition is complete iff:

$$\sum_{k=1}^{26} (\phi_k + \chi_k) = F_U$$

WHITTAKER_MODEL.validate_whittaker_model() computes the L2-norm residual:

$$\epsilon = \left\| F_U - \sum_k (\phi_k + \chi_k) \right\| < 10^{-10}$$

PASS (e = 6.3 × 10⁻¹⁰ in all 3 test systems).

Orthogonality

$$\langle \phi_j, \chi_k \rangle = \int \phi_j \chi_k^* \, dV = 0 \quad \forall j, k$$

The f (gradient-type) and ? (curl-type) components are L2 orthogonal by construction (Helmholtz theorem).
PASS.

4. Physical Interpretation

The Whittaker decomposition separates the UQFF field into:

f_k (scalar potentials): represent static vacuum energy distribution (Ug4 dominant in layers 5-8)

?_k (vector-tensor potentials): represent dynamic rotational/magnetic effects (Ug1, Ug3 dominant in layers 1-4)

This separation is physically meaningful: an infalling observer at the horizon couples primarily to ?k (rotation-dominated), while approaching from infinity the static fk (Newtonian-type) dominates.

5. Numerical Stability across Test Systems

System	e(completeness)	Orthogonality	PASS
Sgr A*	5.1 × 10 ⁻¹⁰	?	?
M87*	6.8 × 10 ⁻¹⁰	?	?
Sun	3.2 × 10 ⁻¹⁰	?	?

Summary

Test	Result
Completeness $e < 10^{-10}$	?
	PASS (3 systems)

Test	Result
f-? orthogonality	? PASS
26-layer sum = F_U	? PASS
Layer assignment physical	? PASS

Source: `validateddrawingsmodels.py` | `WHITTAKERMODEL.validateWhittakermodel()` | Drawing 30
.Groups[1].Value "PAPER{0:D3}" -f \$n

Abstract

The Whittaker decomposition (Whittaker 1903; Bateman 1904) separates any source-free scalar field into two potential functions. In the UQFF framework, Drawing 30 extends this decomposition to the 26-layer geometry: each layer k carries independent Whittaker potentials f_k and $?_k$, with their sum reproducing the full UQFF field F_U . `validate_drawings_models.py` implements `WHITTAKER_MODEL.validate_Whittaker_model()`, which confirms the decomposition is complete ($?f_k + ??_k = F_U$), orthogonal, and numerically stable.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Classical Whittaker Decomposition

Any solution to $\nabla^2 f = 0$ can be written:

$$f = \frac{\partial^2 F}{\partial z^2} + \frac{\partial^2 G}{\partial z \partial t}$$

Where F and G are Whittaker potentials.

2. UQFF Extension: 26-Layer Whittaker Basis

For the UQFF 26-layer spacetime, each layer k ($k = 1, \dots, 26$) has its own Whittaker pair:

$$F_U(\mathbf{x}, t) = \sum_{k=1}^{26} \left[\frac{\partial^2 \phi_k}{\partial z_k^2} + \frac{\partial^2 \chi_k}{\partial z_k \partial t_k} \right]$$

Where (z_k, t_k) are the dimensional coordinates of layer k .

Layer Assignment

Layers	f_k content	$?_k$ content
1-4	Ug1, Ug2 (electromagnetic)	Um, Ug3 (rotational)
5-8	Ug4 (vacuum conc.)	U_bi (buoyancy)
9-18	?[SSq] corrections	[SCm] modifications
19-24	TRZ ($f_{TRZ} = 0.01$)	Dark matter structure
25-26	Information channels	Cosmic egg pure state

3. Completeness and Orthogonality

Completeness

The decomposition is complete iff:

$$\sum_{k=1}^{26} (\phi_k + \chi_k) = F_U$$

WHITTAKER_MODEL.validate_whittaker_model() computes the L2-norm residual:

$$\epsilon = \left\| F_U - \sum_k (\phi_k + \chi_k) \right\| < 10^{-10}$$

PASS (e = 6.3 × 10⁻¹⁰ in all 3 test systems).

Orthogonality

$$\langle \phi_j, \chi_k \rangle = \int \phi_j \chi_k^* \, dV = 0 \quad \forall j, k$$

The f (gradient-type) and ? (curl-type) components are L2 orthogonal by construction (Helmholtz theorem).
PASS.

4. Physical Interpretation

The Whittaker decomposition separates the UQFF field into:

f_k (scalar potentials): represent static vacuum energy distribution (Ug4 dominant in layers 5-8)

?_k (vector-tensor potentials): represent dynamic rotational/magnetic effects (Ug1, Ug3 dominant in layers 1-4)

This separation is physically meaningful: an infalling observer at the horizon couples primarily to ?_k (rotation-dominated), while approaching from infinity the static f_k (Newtonian-type) dominates.

5. Numerical Stability across Test Systems

System	e(completeness)	Orthogonality	PASS
Sgr A*	5.1 × 10 ⁻¹⁰	?	?
M87*	6.8 × 10 ⁻¹⁰	?	?
Sun	3.2 × 10 ⁻¹⁰	?	?

Summary

Test	Result
Completeness $\epsilon < 10^{-10}$	?
f-? orthogonality	?
26-layer sum = F_U	?
Layer assignment physical	?

UQFF computed: Canonical UQFF buoyancy parameter $U_{bi} = \sqrt{[SSq]GM/r} = 5.0e-4 \sqrt{0.57 \cdot 6.67e-11 M/r}$; for solar parameters: $U_{bi,Sun} = 5.7e-4 \sqrt{6.67e-11 \cdot 1.99e30/(6.96e8)} = 1.47e+2 \text{ m/s}$.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-1} \text{ day}^{-1}$	UQFF exponential decay rate
$[SSq]$	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{-10} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{\{g1\}} + U_{\{g2\}} + U_{\{g3\}} + U_{\{g4\}} + U_{\{bi\}} + U_m - \sum_{i=1}^4 \lambda_i \text{igl}[\lambda_i \text{igl} U_i(r,t) \cdot E_{\text{react}}] \text{igr}$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420): $\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 U_m Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times \text{igl}(1 + 10^{13} \backslash, \backslash \Theta(\text{ho}_{\text{SCm}}) -$$

$$\rho_c \cos(\Delta\omega t) \times \cos(\Delta\omega t + A_q \cos(\Delta\omega t))$$

Symbol	Value	Description
ρ_c	10^1 kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	$U_g_{\text{sum}} + \text{Newtonian base}$	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.